# Lecture 38: Deep Learning (Part-3) — ANN Regression & Multi-Class Classification in PyTorch

Welcome to **Lecture 38** of the Machine Learning & Deep Learning Curriculum. This comprehensive module marks the practical transition into applied Deep Learning using **PyTorch**, implementing two complete industrial end-to-end projects:

1. **Continuous Regression**: Predicting Combined Cycle Power Plant electrical energy output ($MW$) from sensor telemetry.
2. **Multi-Class Tabular Classification**: Classifying 7 date fruit cultivars from 34 morphological, geometric, and color features.

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## 1. Master Curriculum & Submodule Directory

| Submodule | Focus Area | Key Concepts Covered | Hand-Drawn Educational Diagram |
|---|---|---|---|
| Notes_38.01 | **ANN for Regression: Problem Formulation** | Power plant thermodynamics, continuous mapping, regression architecture ($4 \to 64 \to 32 \to 1$), parameter count derivation | `ann_regression_architecture.png` |
| Notes_38.02 | **Data Loading & Preprocessing** | `powerplant_data.csv`, `StandardScaler`, `fit_transform` vs `transform`, avoiding data leakage, float32 conversion | `regression_preprocessing_pipeline.png` |
| Notes_38.03 | **TensorDataset & DataLoader** | Tensor pairing, mini-batching ($B = 32$), shuffling mechanics, memory efficiency, multi-worker streaming | `tensordataset_dataloader_pipeline.png` |
| Notes_38.04 | **PyTorch Model Construction** | Subclassing `nn.Module`, `nn.Sequential`, `nn.Linear`, `nn.ReLU`, linear output projection, 2,433 parameter count | `pytorch_nn_module_anatomy.png` |

| Submodule | Focus Area | Key Concepts Covered | Hand-Drawn Educational Diagram |
|---|---|---|---|
| Notes 38.05 | **Training & Validation Loop** | Optimization pillars: `zero_grad()`, `forward()`, `MSELoss()`, `backward()`, `step()`, dual train/val tracking | `pytorch_training_validation_loop.png` |
| Notes 38.06 | **Model Checkpointing & `state_dict`** | Overfitting mitigation, `model.state_dict()`, dictionary serialization, `best_val_loss` thresholding, weight reloading | `model_checkpointing_workflow.png` |
| Notes 38.07 | **Regression Model Evaluation** | `model.eval()`, `torch.no_grad()`, MSE ($17.81$), RMSE ($4.22$ MW), $R^2$ Score ($0.937$), actual vs predicted frame | `regression_evaluation_metrics.png` |
| Notes 38.08 | **ANN for Classification: Formulation** | Date fruit dataset (898 samples, 34 features, 7 classes), output layer sizing ($C = 7$), powers of 2 architecture | `ann_classification_architecture.png` |
| Notes 38.09 | **Multi-Class Preprocessing Pipeline** | `LabelEncoder` ($0 \ldots 6$), feature scaling, `torch.long` target dtypes, `shuffle=True` necessity | `classification_preprocessing_pipeline.png` |
| Notes 38.10 | **Classification Model & Evaluation** | Architecture ($34 \to 64 \to 64 \to 7$), why NO Softmax in model, `CrossEntropyLoss`, `torch.max` argmax, $93.8\%$ accuracy | `cross_entropy_accuracy_evaluation.png` |

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## 2. Global Architecture Comparison

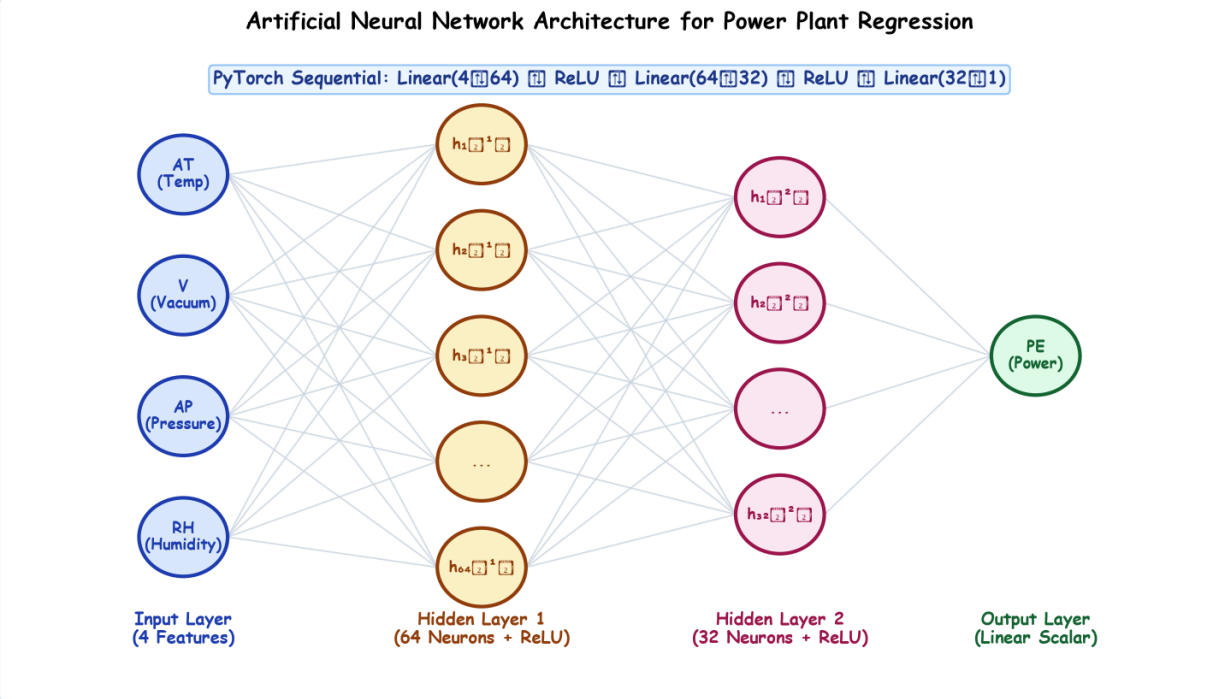
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|                                    ANN ARCHITECTURE COMPARISON
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| REGRESSION ANN (Power Plant)                        | MULTI-CLASS CLASSIFICATION ANN (Date Frui
+-----------------------------------------------------+-----------------------------------
| Target: Hourly Electrical Energy Output (PE)        | Target: 7 Date Fruit Varieties (0 to 6)
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| Input Dimension: 4 Ambient Features                  | Input Dimension: 34 Morphological Feature
| Hidden Layers: [64 (ReLU), 32 (ReLU)]                | Hidden Layers: [64 (ReLU), 64 (ReLU)]
| Output Layer: 1 Neuron (Linear Identity)             | Output Layer: 7 Neurons (Unnormalized Log
| Loss Function: Mean Squared Error (nn.MSELoss)       | Loss Function: Cross Entropy (nn.CrossEnt
| Target Tensor Dtype: torch.float32                   | Target Tensor Dtype: torch.long
| Total Trainable Parameters: 2,433                    | Total Trainable Parameters: 6,855
| Evaluation Metric: R² Score (0.937), RMSE (4.22 MW)| Evaluation Metric: Accuracy (93.8%)
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## 3. High-Yield Architectural Diagram

Below is the master diagram illustrating the end-to-end regression network architecture mapping 4 ambient sensor readings to continuous electrical energy output:




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## 4. Practice Suite & Notebooks

- **Practice Problems (English)** | **Practice Problems (Hinglish)**
- **Full Step-by-Step Solutions (English)** | **Full Step-by-Step Solutions (Hinglish)**
- **Central Code Repository**: `lecture_38/Lecture_Code/` containing `ANN_Regression.ipynb`, `ANN_Classification.ipynb`, `powerplant_data.csv`, and `DateFruit_Dataset.csv`.